

Brenton Stevenson

SKILLS AND TECH

- **Programming / Scripting:** Python, C++, JavaScript, Shell scripting
- **Systems / Tools:** Linux, Git, SSH, Tailscale, Docker, PostgreSQL, Raspberry Pi, NVIDIA Jetson
- **Frameworks / Libraries:** FastAPI, CesiumJS, PyTorch
- **Concepts:** Machine learning, Federated learning, System design, Embedded systems, Data Engineering

EXPERIENCE

FGCU EagleCyberNest Lab **Undergraduate Research Assistant** **Aug 2025 — Present**

- Led experiments for an edge-based federated learning study on EKG anomaly detection and co-wrote the resulting paper, accepted to an IEEE SEC 2026 workshop
- Set up a Raspberry Pi 5 and NVIDIA Jetson Orin Nano edge testbed for on-device model training
- Developing a Mamba-3-based intrusion detection system for healthcare IoMT networks, building on earlier ML-based ransomware detection work

Lee Health **Virtual Health Intern** **Jun 2026 — Present**

- Deploy and configure Caregility virtual care cameras across patient rooms for the Room of the Future initiative, coordinating handoff to construction and plant operations
- Cable-manage completed patient rooms and program in-room TVs for virtual care
- Troubleshoot telehealth carts and other virtual health equipment

FGCU Housing **Resident Assistant** **Aug 2024 — Jun 2026**

- Fostered a welcoming and safe community as a leader in campus residence halls

Plato's Closet **Sales Associate / Buyer / Keyholder** **Sep 2021 — Aug 2024**

- Assisted customers, managed inventory, and completed managerial duties supporting store operations

PUBLICATIONS

Edge-Native Federated Time-Series Anomaly Detection for Resource-Constrained IoMT

- Edge Intelligence Workshop at IEEE SEC 2026 (accepted, to appear Oct 2026)

PROJECTS

- **Edge-Based Federated Learning for EKG Anomaly Detection:** Trained and benchmarked anomaly detection models on the MIT-BIH Arrhythmia dataset; soldered header pins for AD8232 ECG, ADS1115 ADC, and MAX30102 pulse oximetry sensors on a Raspberry Pi 5 client; ran federated training across Raspberry Pi 5 and NVIDIA Jetson Orin Nano nodes (Python, PyTorch)
- **Class Projects:** Led a team building a digital twin prototype for road infrastructure (PostgreSQL, FastAPI, CesiumJS, OpenStreetMap data); autonomous robot vehicle (C++ navigation); SRS Document for Healthcare Appointment Booking System (Requirements, Jira, User Stories, SCRUM, UML)
- **Homelab:** Built a server PC running Plex and Docker; configured a Raspberry Pi 5 as a Moonlight game-streaming client; remote access via Tailscale

EDUCATION

Florida Gulf Coast University

Aug 2023 — Expected May 2027

B.S. in Software Engineering, GPA: 3.1